Portfolio



ABOUT ME

Johanne Vigouroux

UX/UI Designer with a Software Engineering Mindset

Engineering student turned designer. I sketch on paper, test, then build the front-end myself.

- **Available for internships**

How I work

The process I followed to design and build PictoComm, an AAC communication app tailored for autistic users and caregivers.

1 Research

Sensory Needs

Benchmark of existing AAC tools & Physical Boards

Cognitive accessibility research

Dual-user personas (child & educator)

2 Plan

Turning needs into a clear structure

User flows

Persistent navigation paths

3 Make

Designing it, then building it

Dark UI prototypes

Java FX/FXML

4 Test

Audit

WCAG 2.2 compliance check

Cognitive load review

ABOUT THE PROJECT

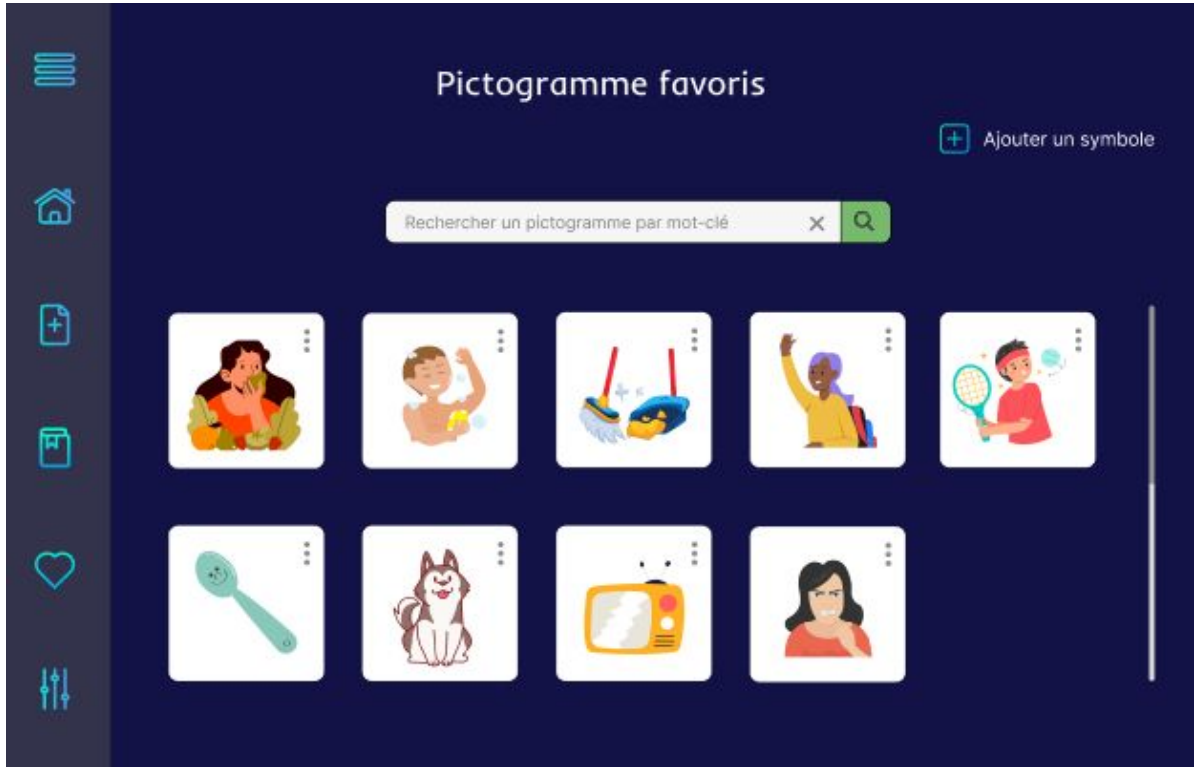
Communication built for specific needs

TIMELINE
3 days, 2024

TEAM
4 students

MY ROLE
Research, UI design, front-end

TOOLS
Figma, ChatGPT, Java FX



WHO IT'S FOR

Autistic learners & caregivers (educators, therapists, parents)

THE PROBLEM

Existing tools cause sensory fatigue, while preparing physical boards takes hours.

WHAT WE BUILT

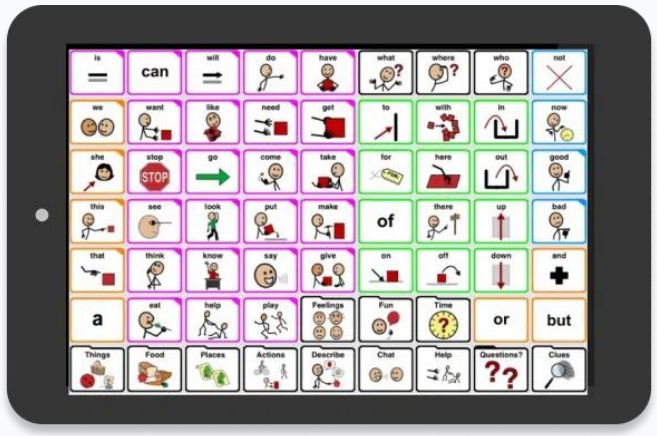
Drag-and-drop routine builder.
WIth WCAG AAA contrast & custom skin tones.

THE PROBLEM

Why standard digital tools fail autistic users

Commercial AAC Apps

Extremely dense grids and steep pricing (200–300€ in a single purchase). Complex nested menus overwhelm autistic users with cognitive friction.



Source: Proloquo2Go App

Generic Routine Apps

Hyper-saturated primary colors, infantilizing animations and zero options to adapt skin tones, borders or text positions.



Source: Avaz AAC App

Physical Boards

Highly tactile and stable but preparation takes hours (printing, cutting, laminating) with constant risk of lost cards.



Translating clinical findings into UI axioms

1. Spatial Predictability

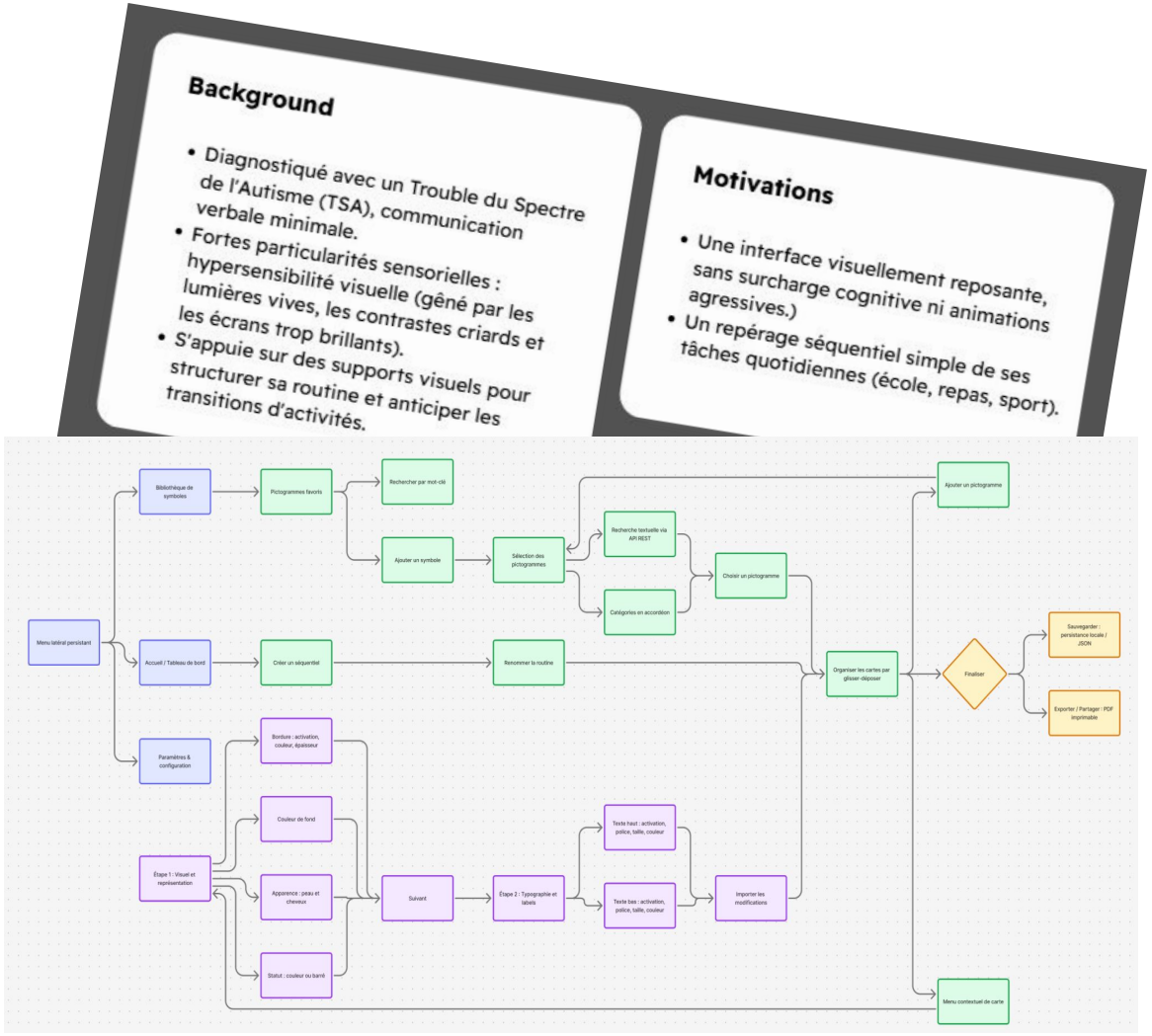
Persistent lateral icons; components must never shift position unannounced.

2. Sensory Soothing

A low-luminance deep navy theme to prevent visual hyper-reactivity.

3. Granular Fine-Tuning

Adjustable border weights, contrast tiers and skin/hair representations.



From research to first sketch: part of the persona and the user flows.

A sensory-friendly dark canvas

1 High Visual Rest

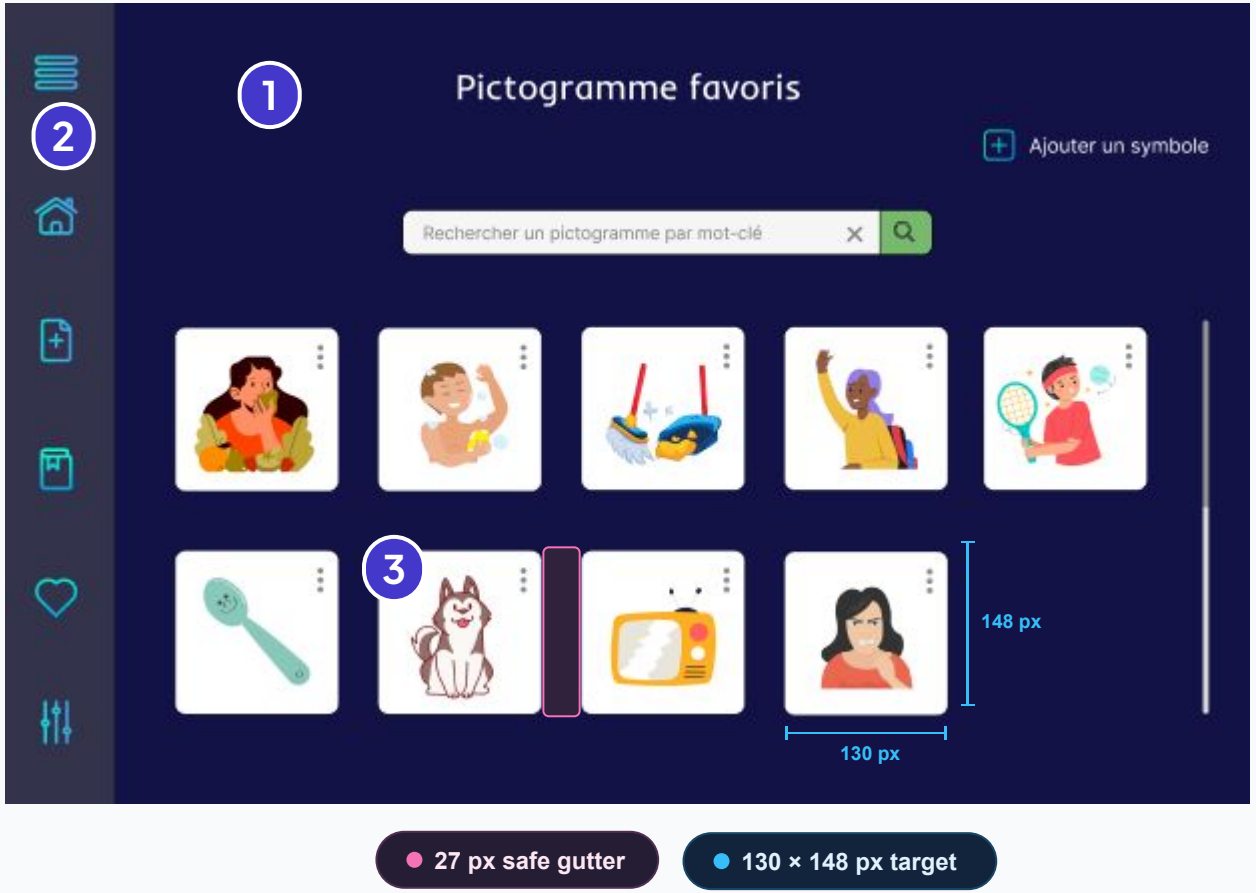
Deep midnight blue (#0d1326) eliminates eye strain and screen glare.

2 Persistent Anchor Bar

Side menu icons maintain stable spatial position to avoid confusion.

3 Touch-Sized Targets

Generous card tiles designed for finger taps (exceeds WCAG AAA 48 px) and safe spacing (exceeds WCAG 2.2 24 px)



Favourite view in the final prototype.

Figma to JavaFX / FXML architecture

1. Design System in Figma

Structured grid, generous spacing tokens, soothing dark UI palette tokens.

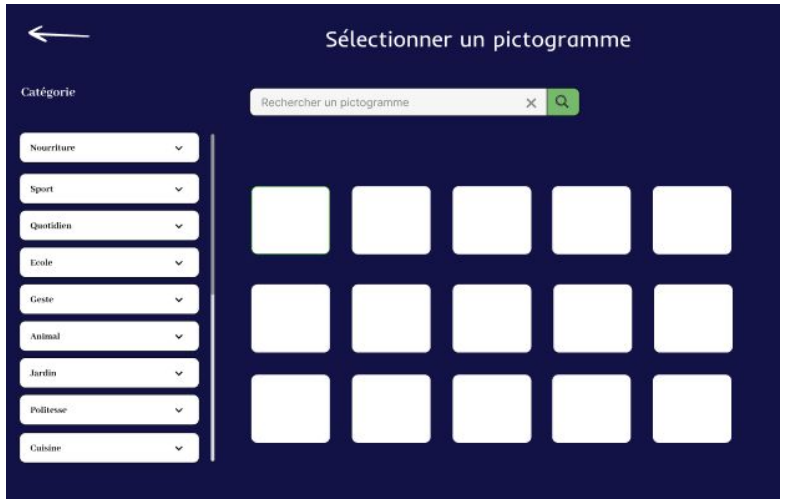
2. JavaFX & FXML Markup

Modular component hierarchy mapping Figma layers directly to FXML.

3. JSON & API Integration

Seamless JSON configuration export and REST API image handling.

Core user journey in 3 steps



1. Search & Pick

Select categorized pictograms or type keyword via API search.



2. Drag into Sequence

Drop tiles in order: breakfast, shower, brushing teeth, clothes.



3. Customize

Adjust border thickness, character skin tone and labels.

Evaluating accessibility & cognitive load

WCAG AAA

Contrast Compliance

Every interactive label and text element passes 7:1 ratio minimum.

0

Sensory Clutter

Eliminated animated distractions, popups and aggressive alerts.

100%

Keyboard Focus

Guaranteed sequential keyboard navigation across all interactive buttons.

RETROSPECTIVE

Key lessons learned

MINDSET

Inclusivity benefits everyone

Designing for cognitive sensitivity creates clearer, more resilient software for all users.

METHOD

Radical simplification

Removing unnecessary decorations protects users from cognitive overload.

RESEARCH

Ethics in accessibility

Scientific research and standards bridge the gap when direct user testing is restricted.

THANK YOU

Let’s talk :)

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AVAILABILITY

3-4 months starting from May

GITHUB

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BASED IN

Rennes, France